



# Politecnico di Torino

## Qubit Electronics

### Lab 6

A. Tudisco, G. Piccinini, F. Riente

14 maggio 2026

#### Lab submission instructions

Create a folder with the following name:

`gr<nn>_lab6`

where  $n$  corresponds to the group number.

In the root of the lab folder, insert the .pdf with the report. *The report's length must be limited to 10 pages.*

At the end of the lab session, create a .zip archive starting from the root folder and name it

`gr<nn>_lab6.zip`

We suggest limiting the number of contents in order to deliver a lightweight file. It is not necessary to upload full output files, as they can be really heavy. If you like to include anything from the output results, provide single images or comments instead.

The created archive must be uploaded by one team member on Portale della Didattica, section Elaborati.

# 1 Introduction

During this laboratory session, we will focus on the energy level splitting in the SiMOS double quantum dot device. It is recommended to run the Python code at home using the server. Here are some useful commands:

- `tmux new -s nameSession`  
to create a new background terminal session called **nameSession**.
- `tmux attach -t nameSession`  
to attach to the background terminal session **nameSession**.
- `tmux kill-session -t nameSession`  
to kill the background terminal session **nameSession**.

To detach from the background terminal session, use:

`ctrl + b + d`

## 2 Python script

Given the provided Python code, add the Schrodinger and Poisson solvers inside the for loop, where indicated. Then, add the required lines of code to calculate the energy difference between the two energy levels (ground state and first excited). Save the results related to the difference and the inter-dot barrier vector.

**Question:** Which parameters related to the structure differ from the previous laboratory?

**Question:** Report the line of code added.

## 3 Analysis of the results

Write a script to perform an exponential fit of the values of the energy level splitting versus the inter-dot barrier and comment on the results obtained. Is the behavior for low inter-dot barrier gate bias voltage as expected? Explain why or why not.